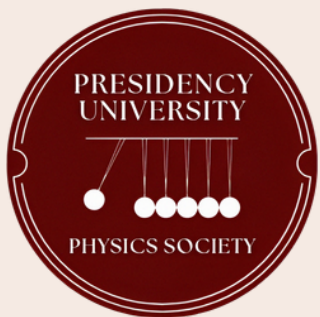




Physics Lecture Series

PHYSICS BEYOND THE CLASSROOM



Dr. Suman Kundu
SISSA, Trieste, Italy

A FEW PUZZLES IN PHYSICS



About the Course

1. **Puzzles leading to EM and STR.** Faraday to Maxwell's work; the Michelson–Morley experiment; Einstein's two postulates and the Lorentz transformation.
2. **Puzzles leading to GR.** Incompatibility of Newtonian gravity with STR; the equivalence principle; why $\eta_{\mu\nu}$ must become a dynamical $g_{\mu\nu}(x)$; the Einstein field equations.
3. **Consequences of GR.** Perihelion precession, light deflection, gravitational redshift (consequence in Modified gravity theory), gravitational waves, black holes, and cosmology.
4. **Need for QFT formalism.** Blackbody radiation; negative probabilities in relativistic QM; pair production and variable particle number; causality violation and its resolution.
5. **Consequences of QFT.** Quantum Electrodynamics and $g - 2$; gauge symmetry (using group theory), observables (experiments in LHC).
6. **One of the current puzzles.** Problem with quantising Gravity; the black hole information paradox; string theory, and holography

REGISTER HERE



<https://forms.gle/n9aWAtSD5vnXVYNK7>

**LAST DATE OF
REGISTRATION: 15TH JUNE**



From 23rd June, 2026



Tuesdays & Saturdays



8.30 PM IST (1 Hour)



Online Mode



6 Classes

REGISTRATION IS OPEN TO ALL

► Registration Fees:

- ₹50 for members of the Presidency University Physics Society
- ₹100 for non-members and Students from Other Institutes



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